

NSCC

Data Science in Business



Plan of Action **PROPOSAL**



Presented by:

Bea Sauve, Elton Nichols, Kainen Osborne

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Executive Summary



Timeless Transport Models has been proudly serving customers on a Business to Business basis. This document outlines the strategic plan to grow profits between June 2005 and December 2005.

Drawing in four integrations of analyses; resource allocation, sales performance, and forecasting. Identifying data-driven recommendations designed to maximize revenue, reduce costs, and capitalize on seasonal demand patterns.

Business Context

Timeless Transport Models is a B2B manufacturer and distributor of modeled vehicles operating three factories across six product lines: Classic Cars, Vintage Cars, Motorcycles, Planes, Ships, Trucks & Buses, and Trains. Since 2003, the company has accumulated production, sales, and distribution data. As the company enters the second half of 2005, leadership requires production forecasts, cost analysis, and strategic profit projections to guide operations through December 2005.

Summary of Recommendations

Recommendation	Focus Area	Expected Impact
1. Halt Train Production	[Factory Production]	Omit Less Profiting Items
2. Floor Sale Prices	[Future Profits]	Increase Profits
3. Recommended Productions	[Production Volume]	Stable Production
4. Email Marketing	[Customer Engagement]	Customer Interaction and Sales

Resource Analysis

This section summarizes the findings from the analysis project. The analysis used a weighted least squares regression model to determine the cost per unit (CPU) for each product line. The model was also used to understand factory level structures.

Regression Model Overview

A weighted regression mode was fit to weekly production data across three factories influenced by outliers. The model Achieved an R^2 of 0.928, indicating that 92.8% of the variance in weekly production cost is explained by production volumes and factory assignment.

Metric	Value
R^2	0.9282
Adjusted R^2	0.9266
Residual Std. Error	226.4
Observations	355 (outliers excluded)

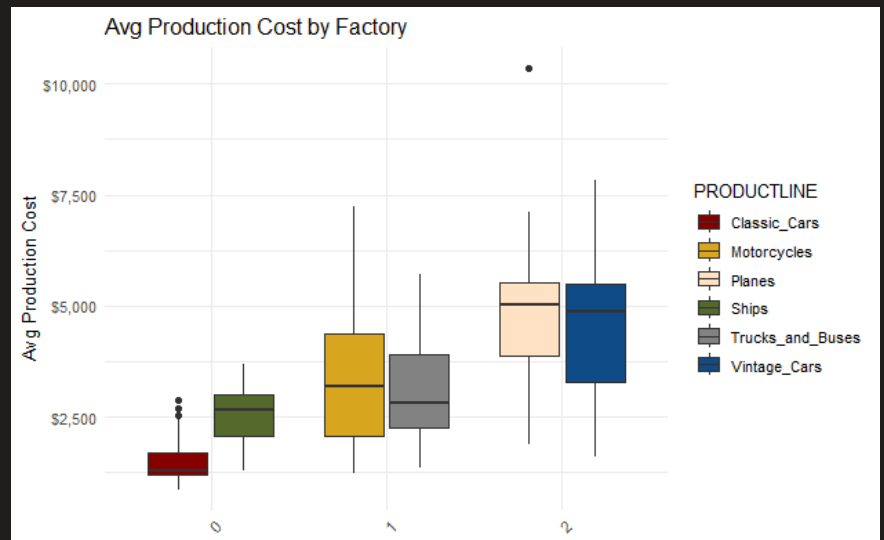
Cost Per Unit (CPU) by Product Line

The table below shows the regression derived CPU for each product line. These values represent the marginal cost of producing one additional unit of each product.

Product Line	CPU (Regression)	Interpretation
Trucks & Buses	\$91.04	Highest cost per unit
Classic Cars	\$83.73	High cost, low volume
Planes	\$83.10	High cost, moderate volume
Motorcycles	\$75.09	Mid-range cost
Vintage Cars	\$66.67	Moderate cost
Ships	\$15.91	Lowest CPU, highest volume

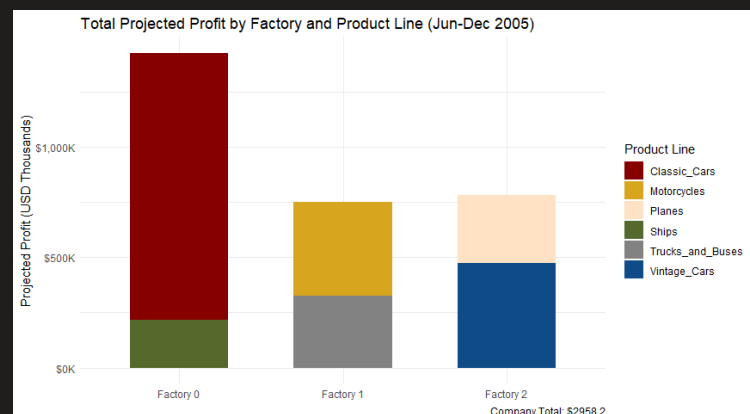
Factory Performance

The regression model identified a base operating cost (intercept) for each factory, representing fixed overhead such as labor and electricity. Factory 0 carries a base cost of \$448.03 per week and serves as the model baseline. Factory 1 operates at a reduced base cost of \$256.69, approximately \$191 less than Factory 0. Although this difference carries a p-value of 0.48, indicating limited statistical significance. Factory 2 has the highest base cost at \$546.77, roughly \$98 more than Factory 0, and also shows the highest average monthly production costs overall (exceeding \$7,000 in some periods). Factory 2 produces Planes and Vintage Cars, two lines with notably different CPUs, and the high cost of Plane production appears to drive up the blended cost for that facility.



Key Takeaways

Classic Cars are the most profitable product line despite relatively high production costs (\$83.73/unit) and low production volumes. Ships have the lowest CPU (\$15.91) and the highest production counts, yet rank among the least profitable due to low selling prices. An inverse relationship was observed between Classic Cars (low production, high sales) and Ships (high production, lower sales), suggesting a significant back stock dynamic that warrants internal review. Instances of items being sold below their regression CPU were also identified across multiple product lines, representing a key risk to margin health. Train production data was absent entirely, and production should be halted until complete records are available.



Recommendations

Recommendation 1: Halt Train Production & Align Production to Seasonality

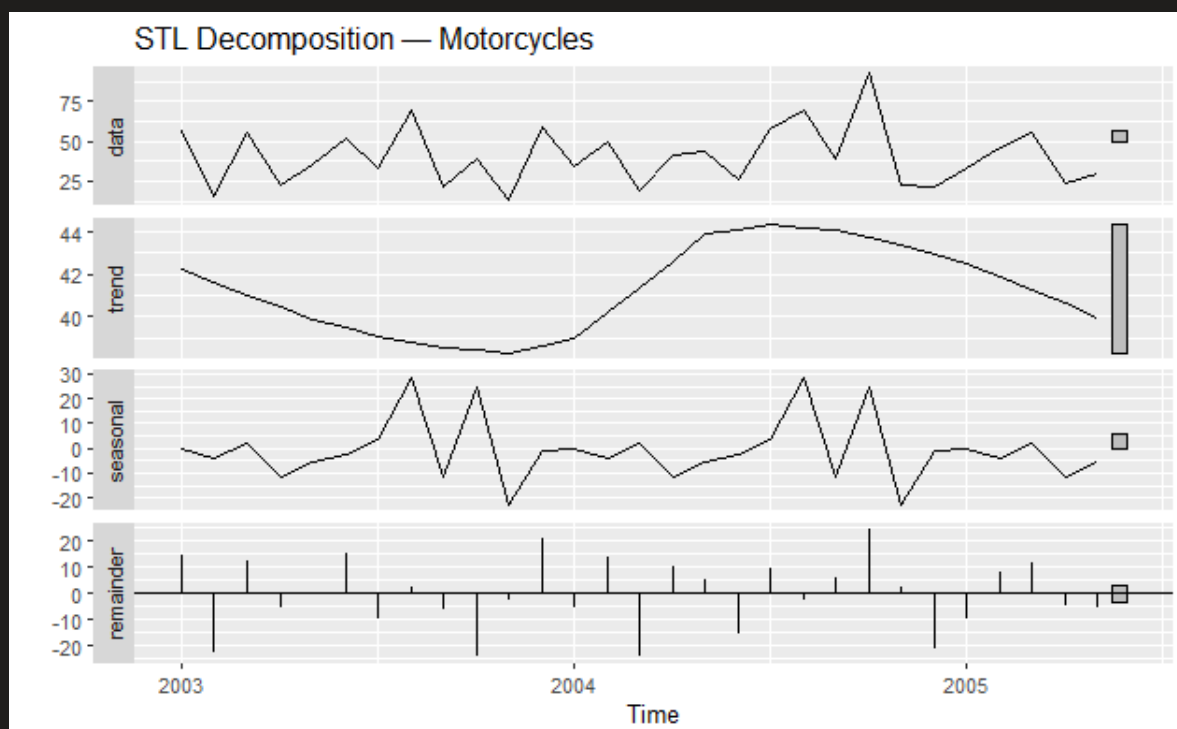
Background

Train production data is entirely absent from the current dataset, making it impossible to plan costs, volumes, or forecasts for this line. Continuing to allocate factory time and resources to Trains without data creates invisible risk. Additionally, all active product lines show clear seasonal production patterns with peaks in August and November–December. Aligning production schedules to these cycles will ensure adequate inventory during high-demand periods

Supporting Data

STL time series decomposition of all active product lines confirmed consistent, repeating seasonal patterns. Production forecasts (June–December 2005) were generated using STL models and reveal expected holiday surges for Motorcycles, Planes, Ships, and Vintage Cars.

See: STL Decomposition plots and Production Forecast charts for more visuals (Project_3_Technical_Report.Rmd)



Metric	Finding
Trains production records	No data available — line cannot be planned or costed
Seasonal peaks identified	August and November–December across most product lines
Expected holiday production surge	Ships: 170+; Planes: 90+; Motorcycles: 60+; Vintage Cars: 85+

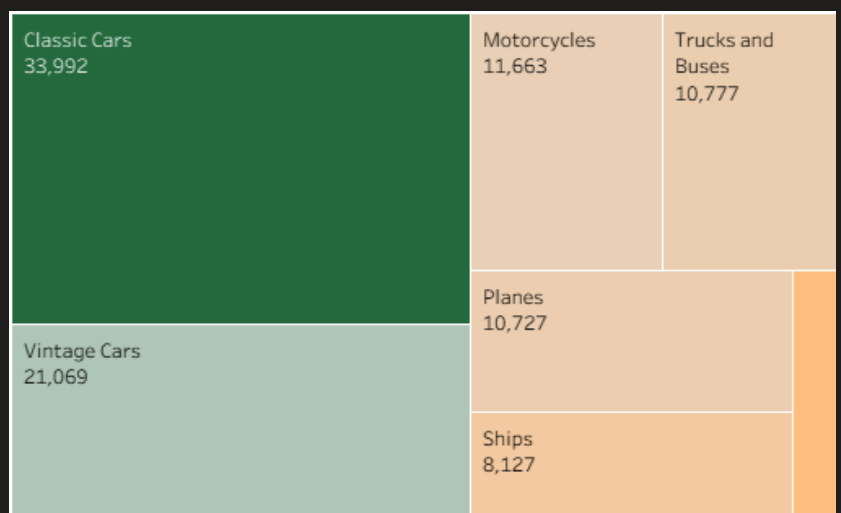
Action Steps

1. Immediately halt all Train production and reallocate factory capacity to active lines.
2. Schedule production increases for August and November–December based on STL forecast ranges.
3. Obtain and audit Train inventory and sales data before any future production decisions.

Expected Impact

Halting Train production eliminates untracked resource expenses. The lack of customer interest purchasing the product furthers reasoning to halt production. Additionally, aligning production to seasonal demand ensures sufficient inventory during the highest-revenue months (Aug, Nov–Dec), reducing the risk of stockouts or over production during low demand periods.

Total Order: Demand for trains is so small it's barely visible on the chart.



Recommendation 2: Enforce a Price Floor Above Production CPU

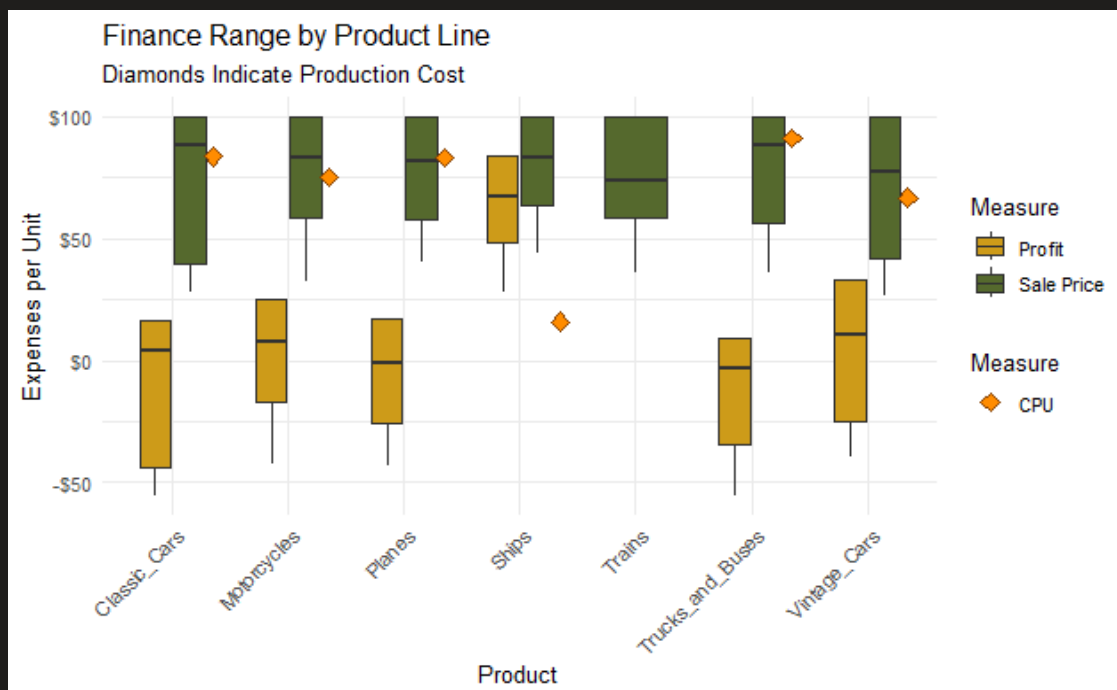
Background

Analysis of product line profits revealed that items across multiple product lines are being sold at prices below their regression estimated cost per unit (CPU). Selling below cost directly deflates profitability. While total profit figures appear strong, this is likely sustained by high order volumes and back logged production. If left unaddressed, this pattern poses a long-term threat to the company's financial health.

Supporting Data

The Finance Range by Product Line box-and-whisker chart showed selling prices dipping below the CPU (orange diamond) for several product lines. The CPU by Product Line bar chart confirms production costs ranging from \$15.91 (Ships) to \$91.04 (Trucks & Buses).

See: Finance Range by Product Line chart and CPU by Product Line bar chart (Project_3_Technical_Report.Rmd)



Metric	Finding
Trucks & Buses CPU	\$91.04 — highest production cost; minimum sale price must exceed this
Ships CPU	\$15.91 — lowest CPU; pricing flexibility exists but floor must still apply
Below-cost sales instances	Observed across multiple product lines per box plot analysis

Action Steps

1. Set a minimum unit sale price for each product line at CPU + a minimum margin buffer (recommended: 10–15%).
2. Audit current open orders and flag any priced below the CPU floor for renegotiation.
3. Update pricing guidelines in the sales system to prevent future below cost transactions.

Expected Impact

Implementing a price floor ensures every unit sold contributes positively to profit. Even a conservative floor of CPU + 10% would meaningfully improve product line margins, especially for high volume lines like Ships.

Recommendation 3: Launch Targeted Regional Email Marketing Campaigns

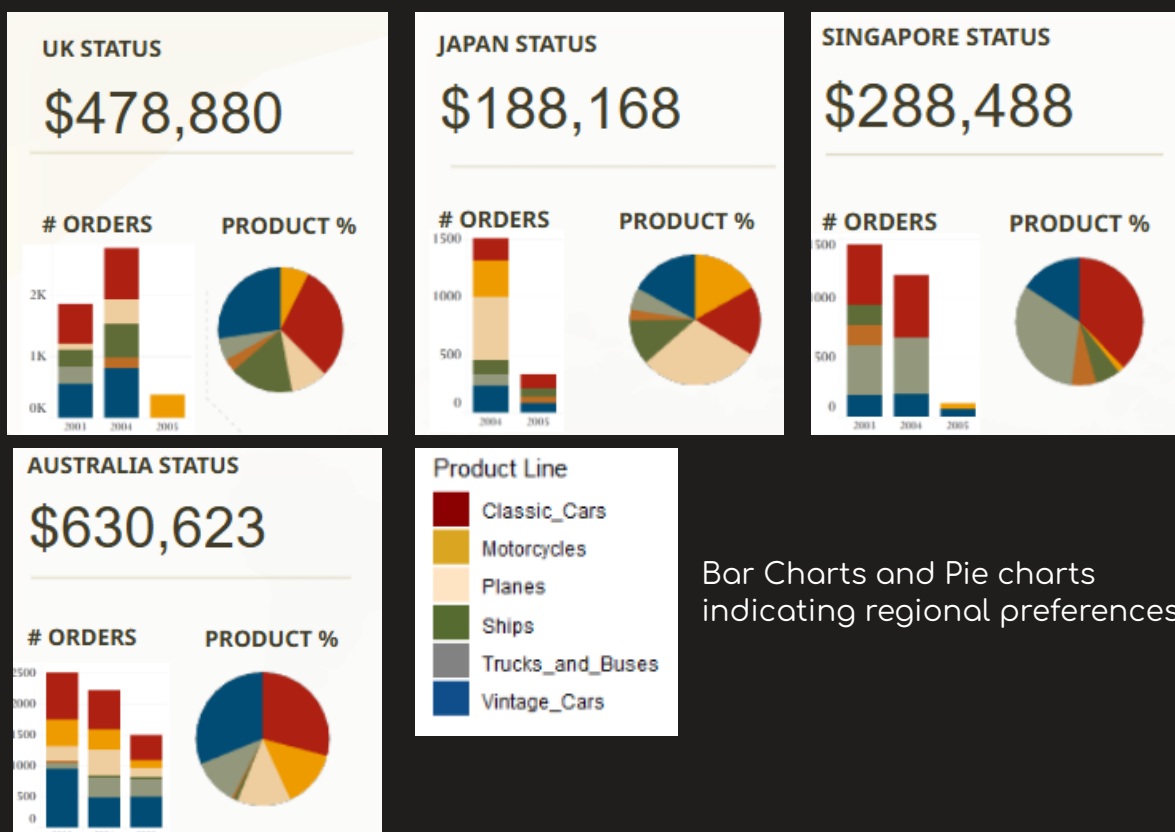
Background

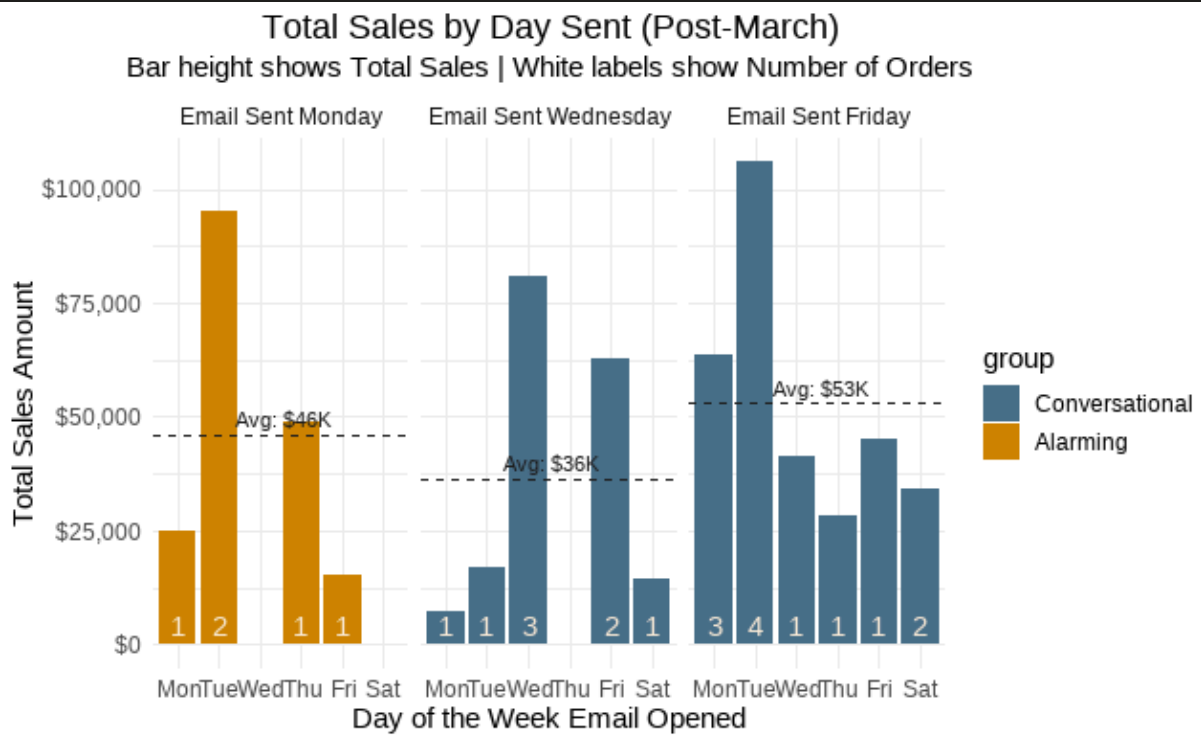
Regional sales data shows that North America, Spain, and France are the top revenue contributors. There resides several high potential markets such as the UK, Japan, Singapore, Australia, and South America. Showing strong preferences for specific product lines that are currently under marketed. Targeted campaigns directed towards regional product preferences, sent on high engagement days (Wednesdays and Fridays), represent an untapped growth opportunity.

Supporting Data

Regional Sales Performance analysis identified product preferences by territory. Top contributors (North America: \$3,627,983; Spain: \$1,215,686; France: \$1,110,916) skew toward Classic Cars, Vintage Cars, and Motorcycles. Emerging markets show distinct preferences: UK favors Planes and Ships; Japan favors Planes and Motorcycles; Singapore favors Trucks & Buses; Australia favors Motorcycles, Trucks, Buses, and Planes.

See: Regional Sales Performance slides (Plan_of_Action_P3.pptx, slides 13–14)





Email Metrics portraying peak customer engagement with conversational emails for emails that are sent on Wednesdays and Fridays.

Metric	Finding
North America total sales	\$3,627,983 — largest market, focus on Classic & Vintage Cars
UK + Japan + Singapore + Australia combined	\$1,585,559 in untapped/emerging revenue; product fit confirmed
Optimal campaign send days	Wednesdays and Fridays (per email engagement analysis)

Action Steps

1. Develop product-specific email campaigns for UK (Planes/Ships), Japan (Planes/Motorcycles), Singapore (Trucks & Buses), and Australia (Motorcycles/Planes/Trucks).
2. Schedule all email campaigns to send on Wednesdays and Fridays for maximizing open and click-through rates.

Expected Impact

Targeted campaigns to the four emerging markets are expected to increase order volumes from regions with less order engagement. If these markets trend toward the average of the top three contributors, the combined revenue is substantial.

Conclusion

This plan of action outlines a data driven strategy for Timeless Transport Models to maximize profitability through the second half of 2005. By drawing on weighted regression modeling, STL time series forecasting, and regional sales analysis, three clear priorities have emerged: streamlining production, protecting margins, and expanding market reach.

Halting Train production removes an untracked resource drain and frees factory capacity for lines with demand. Aligning output to the seasonal peaks in August and November–December ensures inventory is where it needs to be during the company's highest-revenue months. Enforcing a price floor above each product line's regression estimated CPU directly addresses the below cost selling patterns identified in the financial analysis. Finally, launching targeted email campaigns to underpenetrated markets in the UK, Japan, Singapore, Australia, and South America capitalizes on regions that are currently underperforming.

Classic Cars remain the company's most valuable product line by profit margin, and Vintage Cars and Motorcycles show strong demand from top contributing regions. Ships, while the highest volume and lowest costing product to produce, represents an opportunity to improve returns through optimized pricing. Together, these recommendations allow Timeless Transport Models to close 2005 with stronger margins, more efficient operations, and a broader, more engaged customer base.